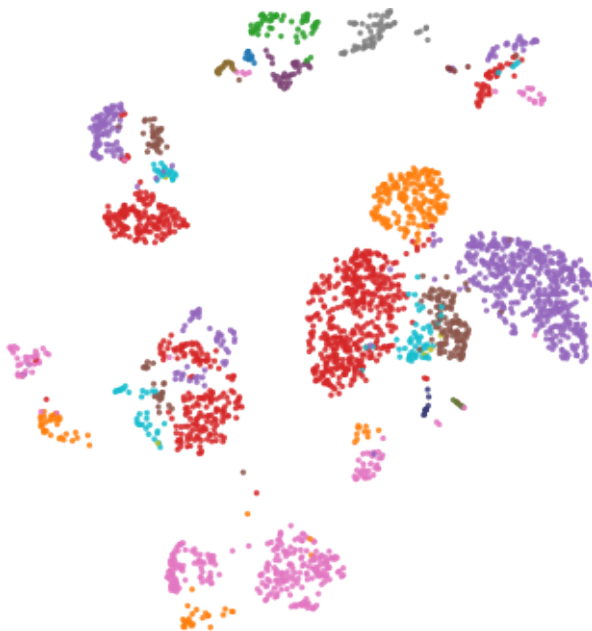
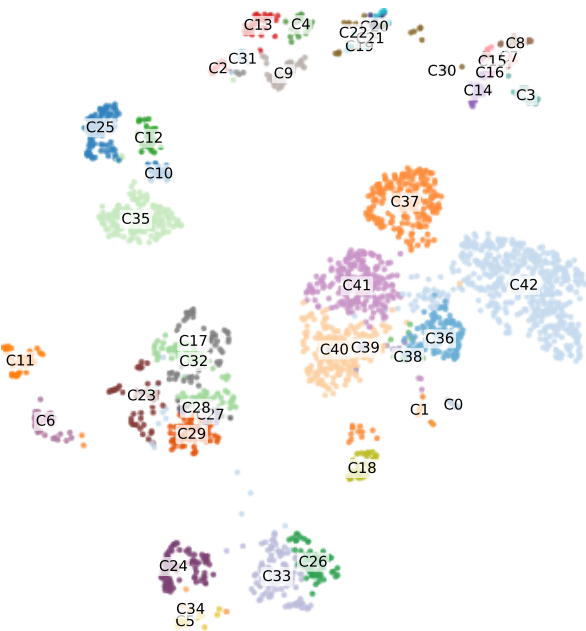


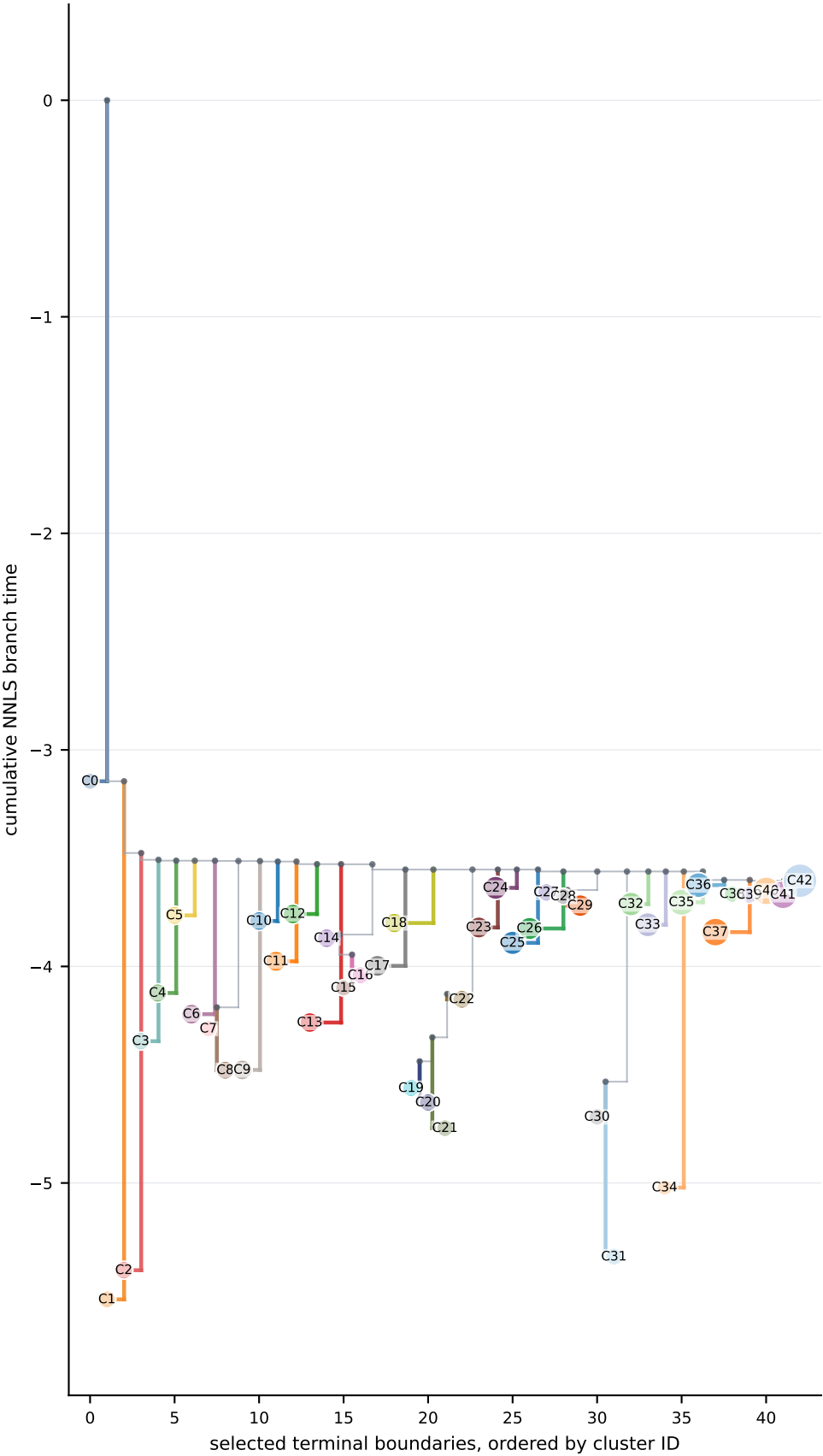
Reference labels on UMAP



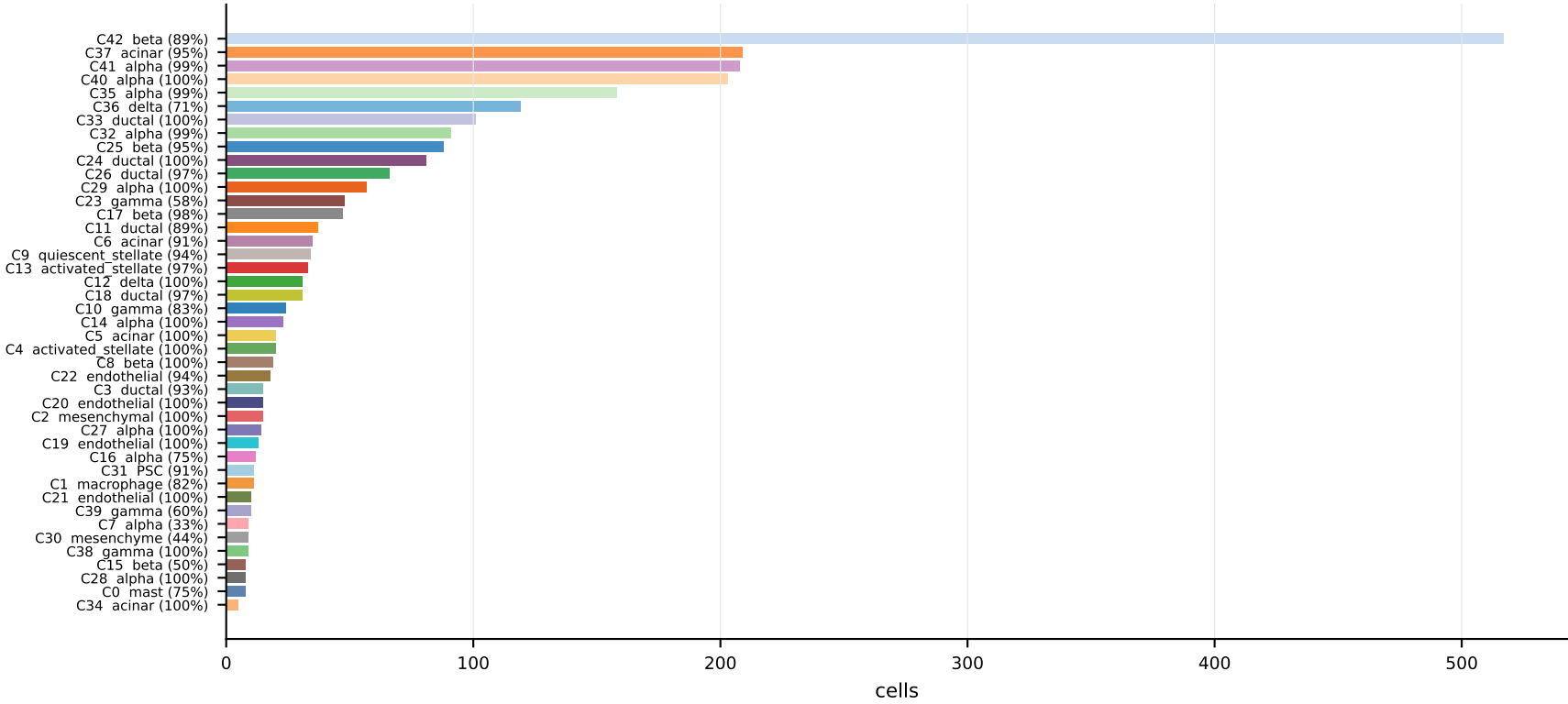
Selected NNLS-TBS clusters on UMAP



Compact selected-boundary tree, same colors

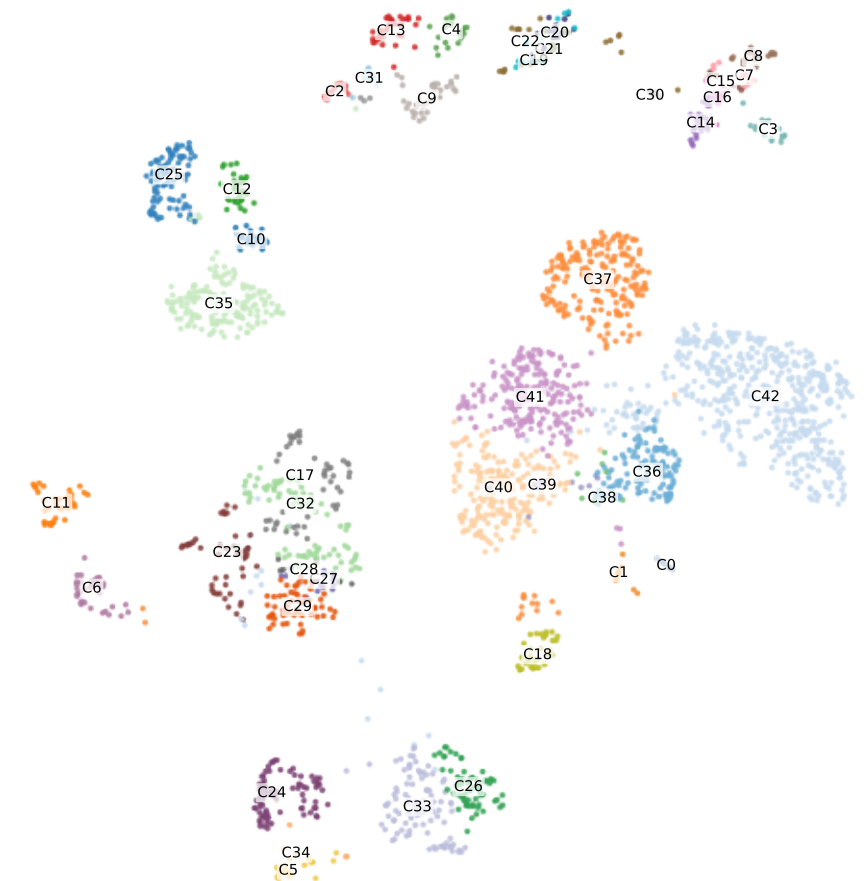


Selected clusters: size and dominant reference label

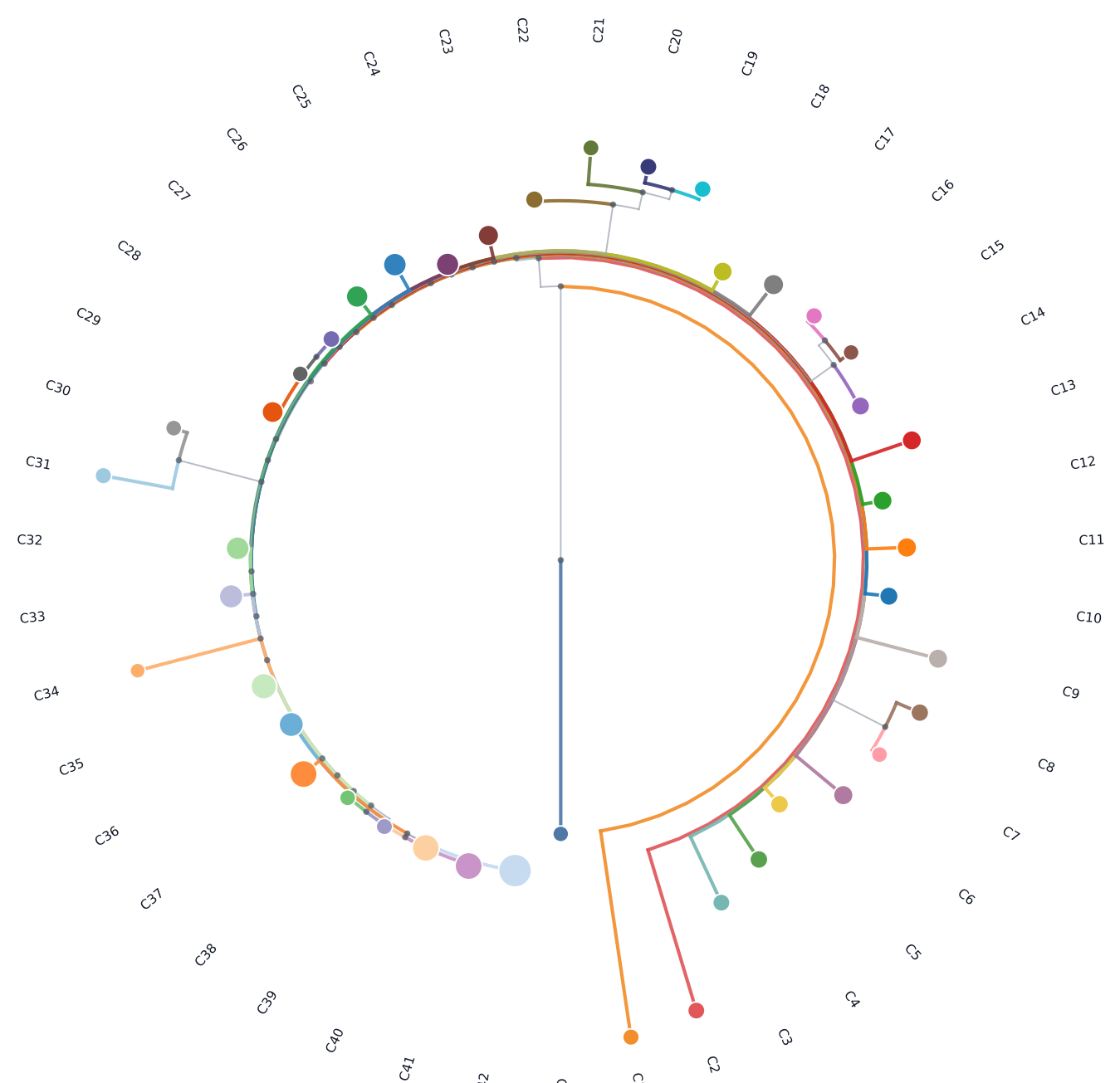


Adult pancreas scRNA benchmark radial tree
TBS adaptive diffusion recomputed-NNLS branch-time (k90, alpha=0.01, edge=0.001)

Selected NNLS-TBS clusters on UMAP



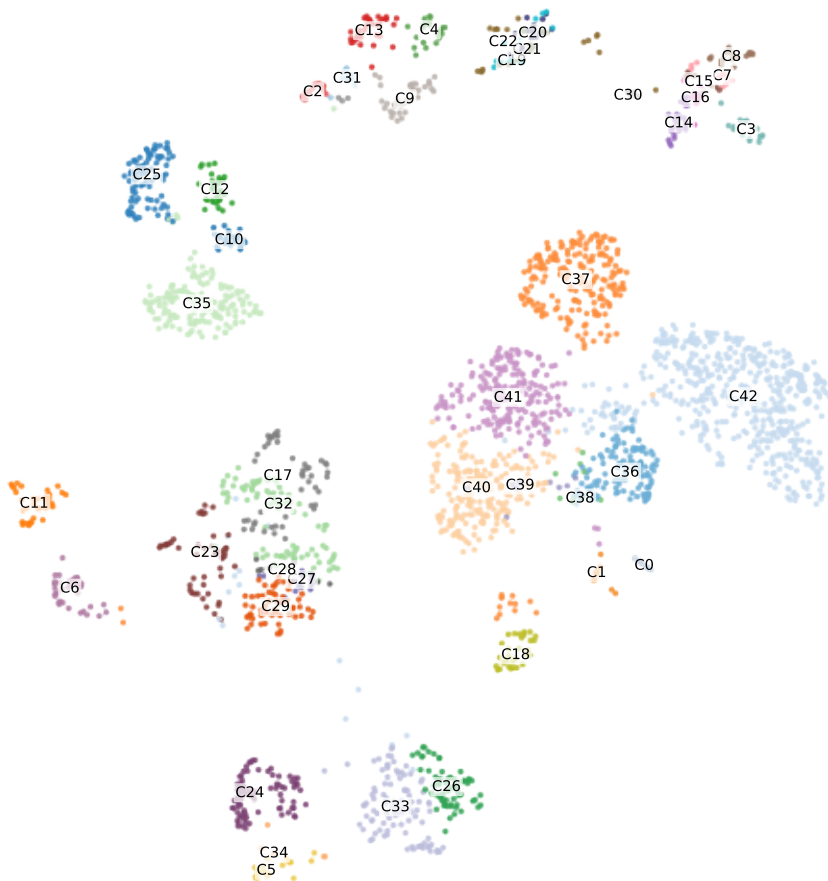
Selected-boundary radial tree, same colors



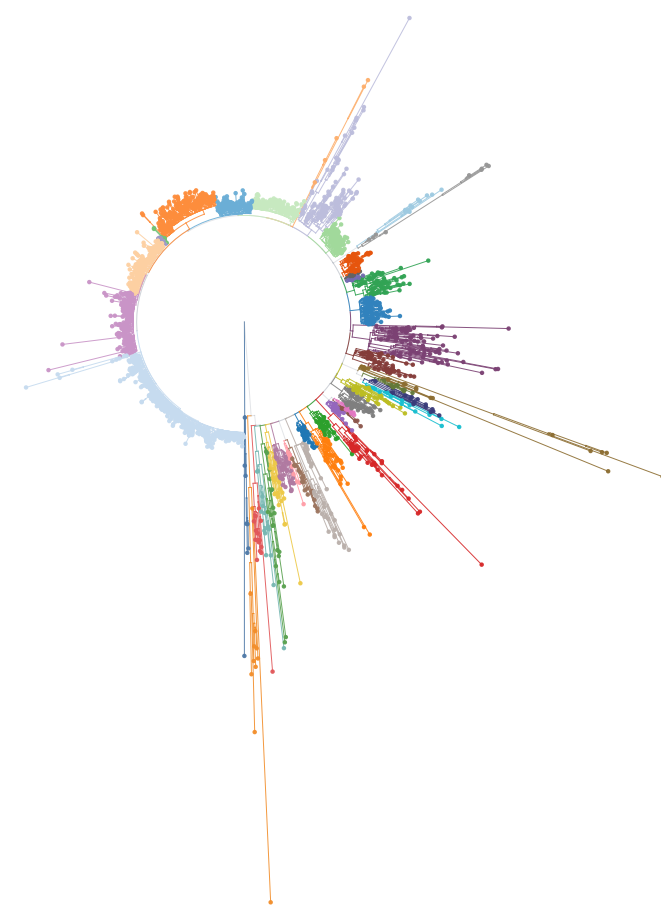
radius = cumulative NNLS branch time; colors = selected cluster IDs

Adult pancreas scRNA benchmark full selected tree
TBS adaptive diffusion recomputed-NNLS branch-time (k90, alpha=0.01, edge=0.001)

Selected NNLS-TBS clusters on UMAP



Real full radial tree, same leaf colors

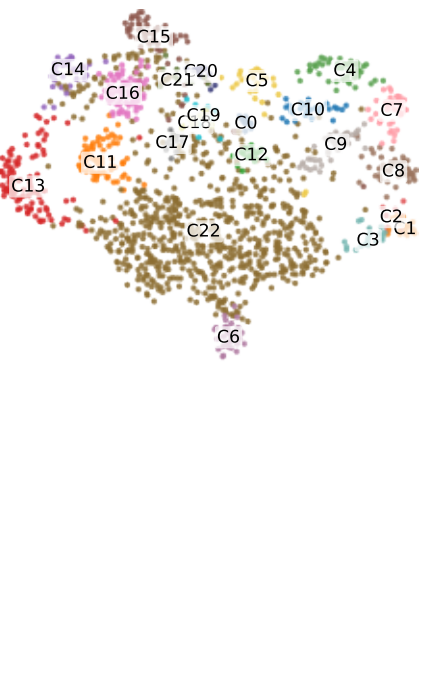


all 2,500 leaves; radius = cumulative NNLS branch time; colors match the labeled UMAP

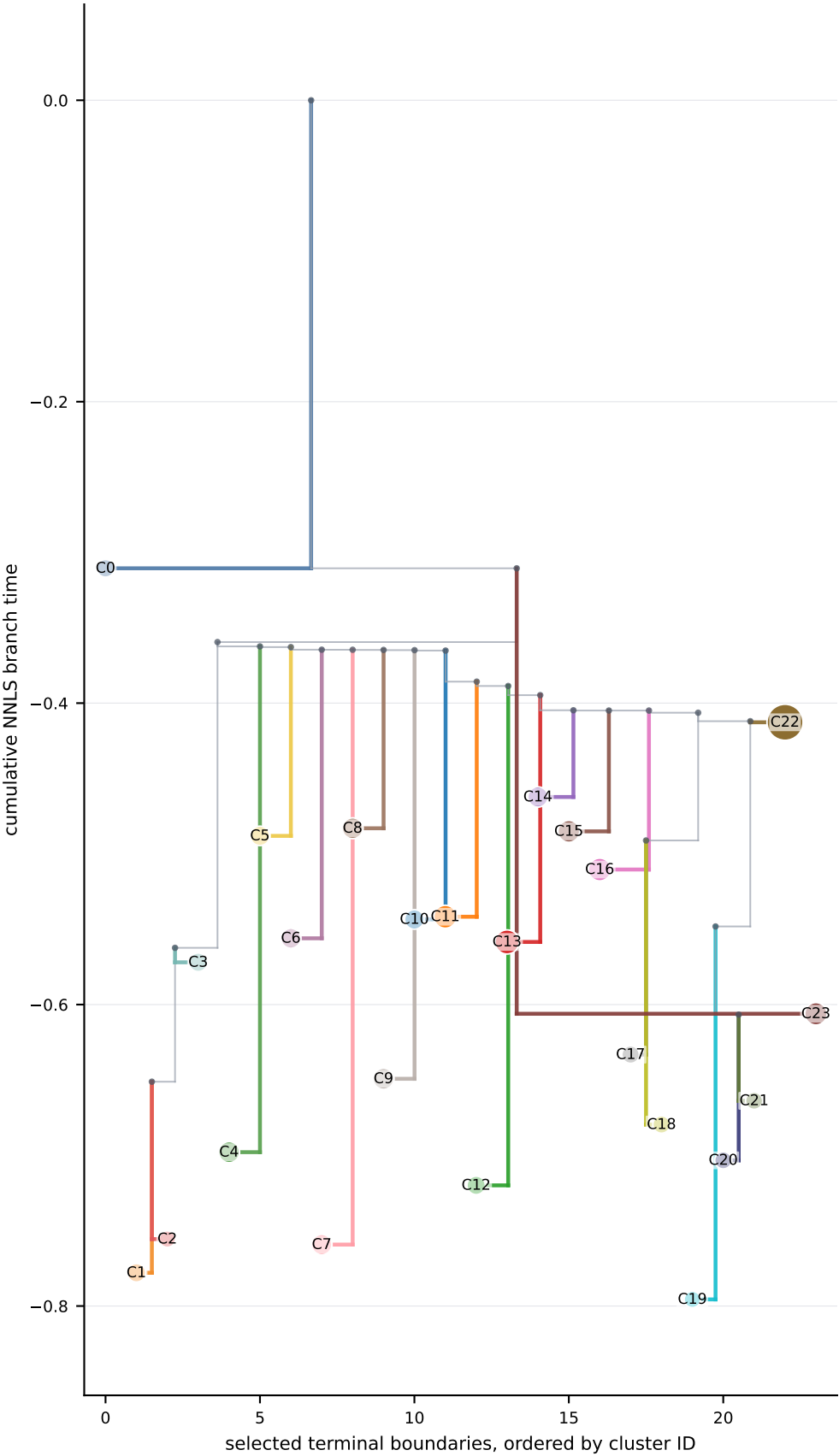
Reference labels on UMAP



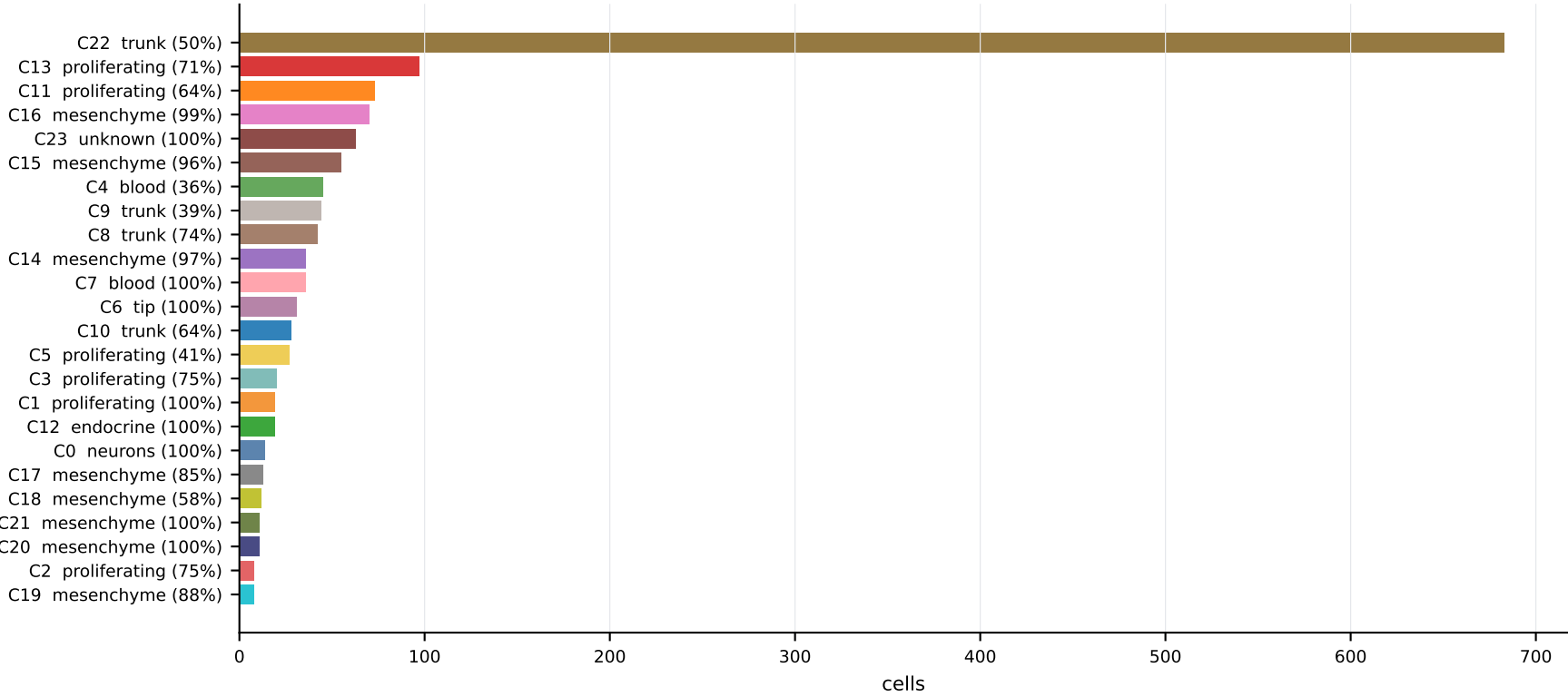
Selected NNLS-TBS clusters on UMAP



Compact selected-boundary tree, same colors

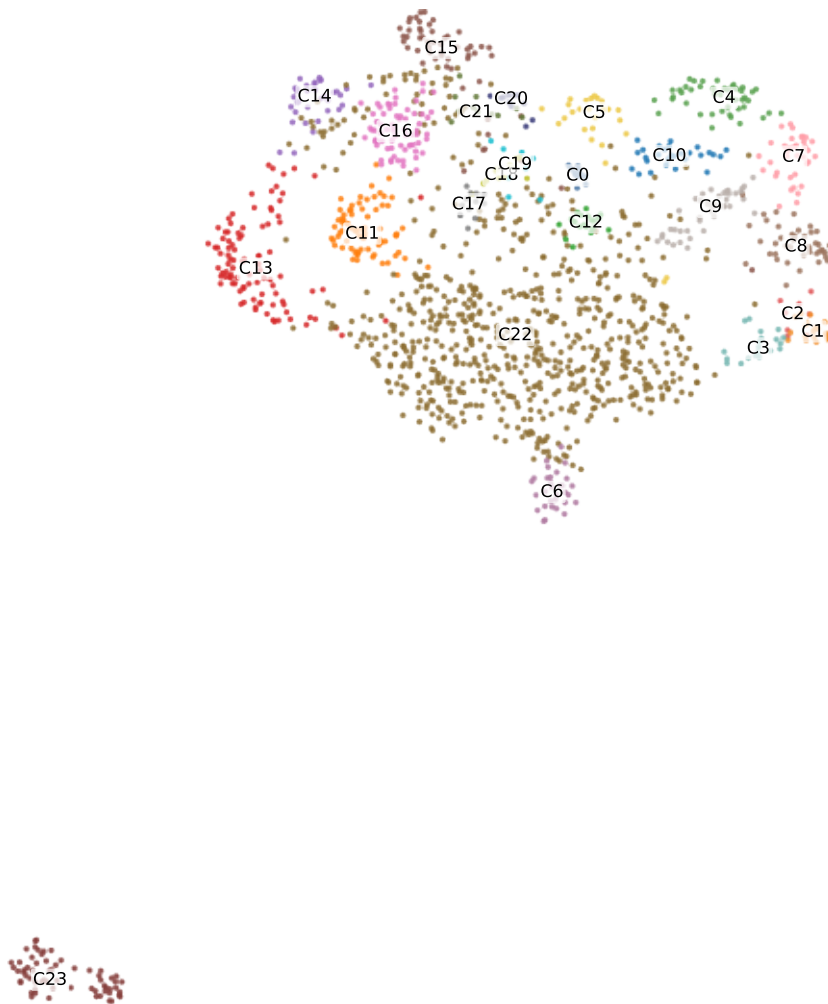


Selected clusters: size and dominant reference label

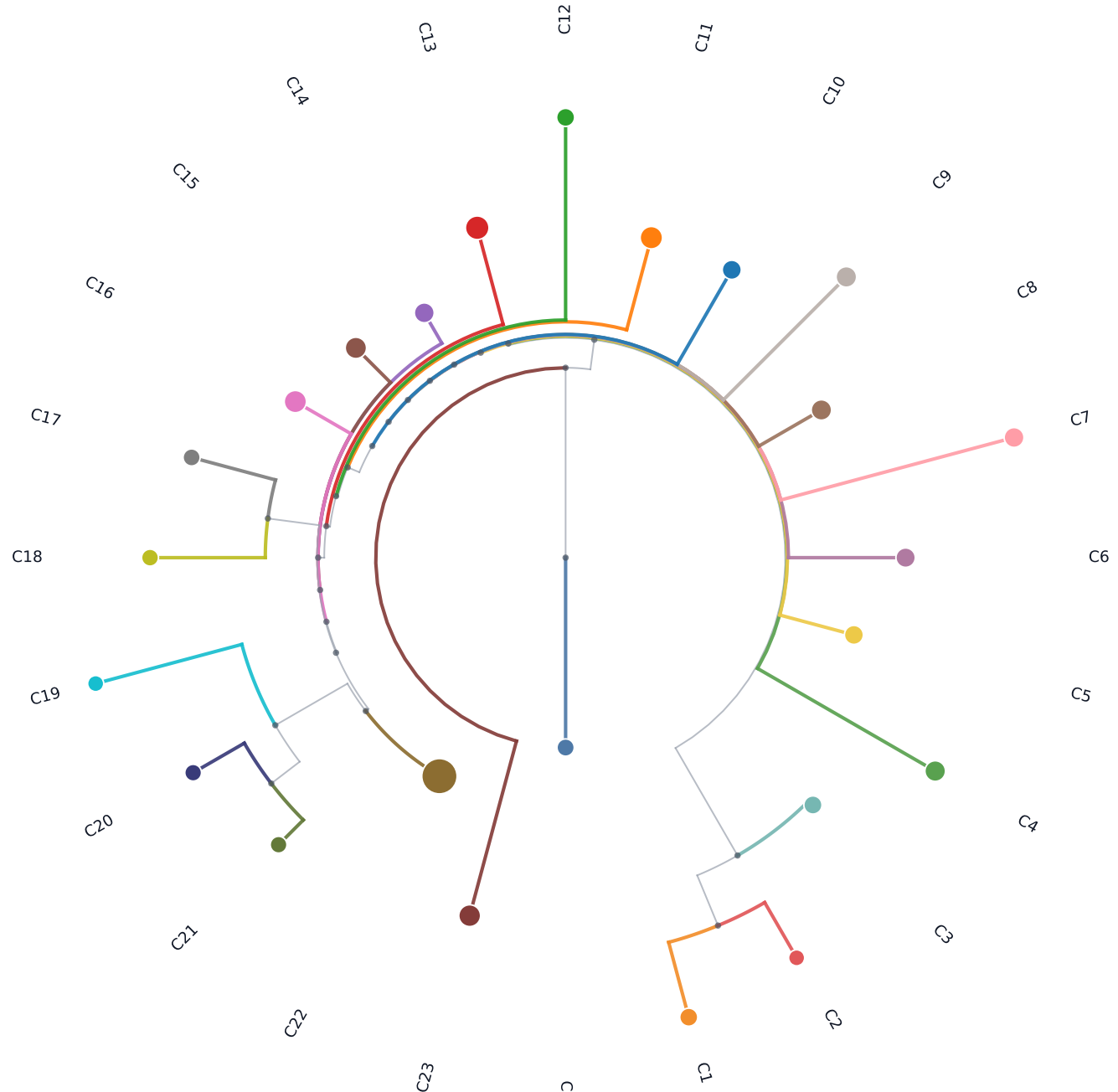


Goncalves fetal pancreas progenitor benchmark radial tree
TBS adaptive diffusion recomputed-NNLS branch-time (k90, alpha=0.01, edge=0.001)

Selected NNLS-TBS clusters on UMAP



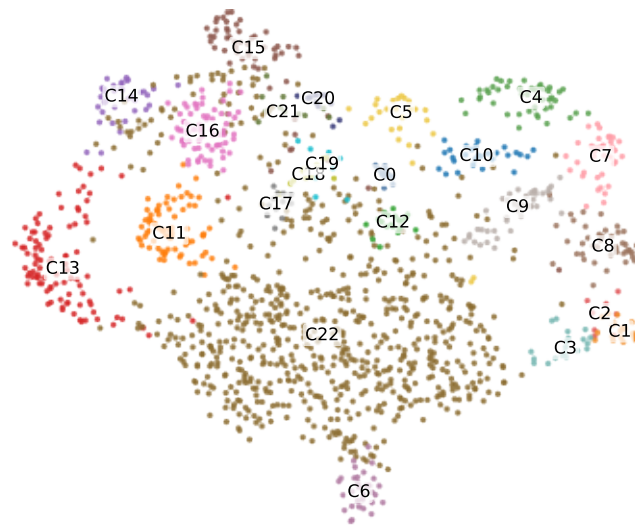
Selected-boundary radial tree, same colors



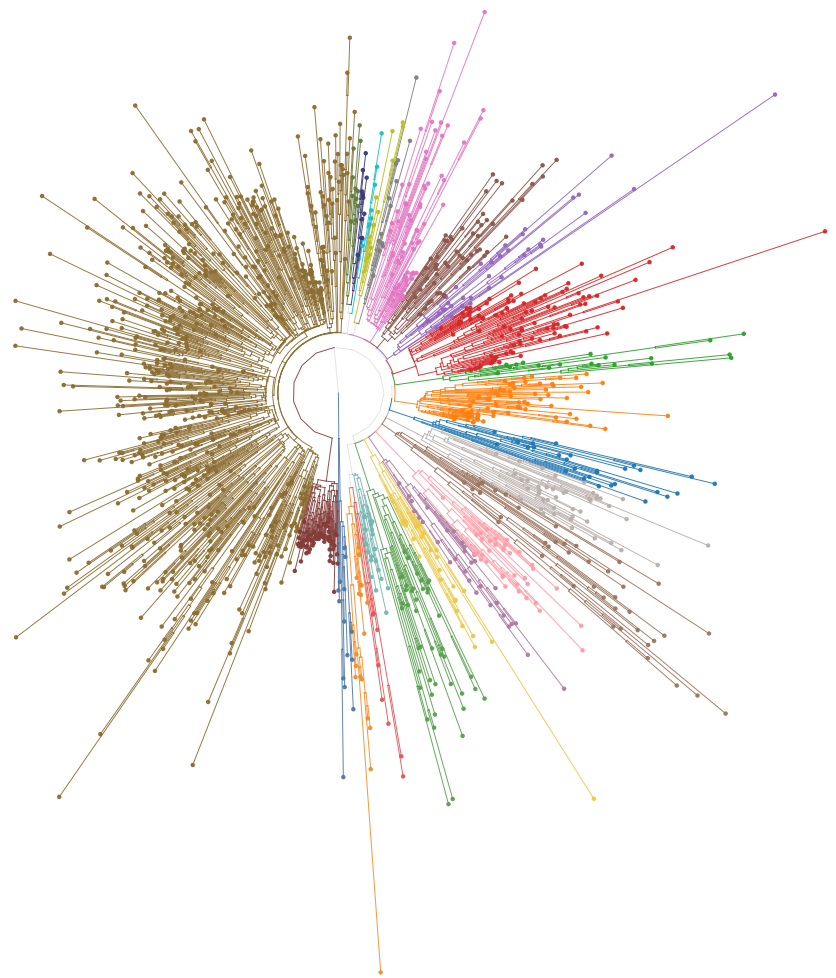
radius = cumulative NNLS branch time; colors = selected cluster IDs

Goncalves fetal pancreas progenitor benchmark full selected tree
TBS adaptive diffusion recomputed-NNLS branch-time (k90, alpha=0.01, edge=0.001)

Selected NNLS-TBS clusters on UMAP



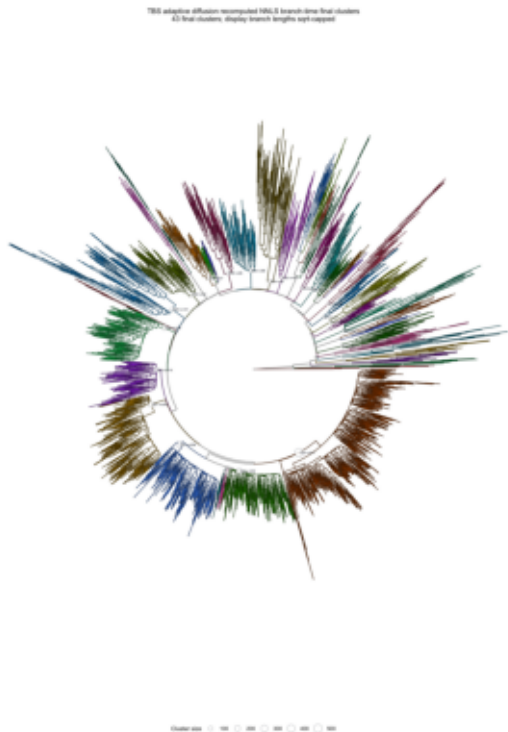
Real full radial tree, same leaf colors



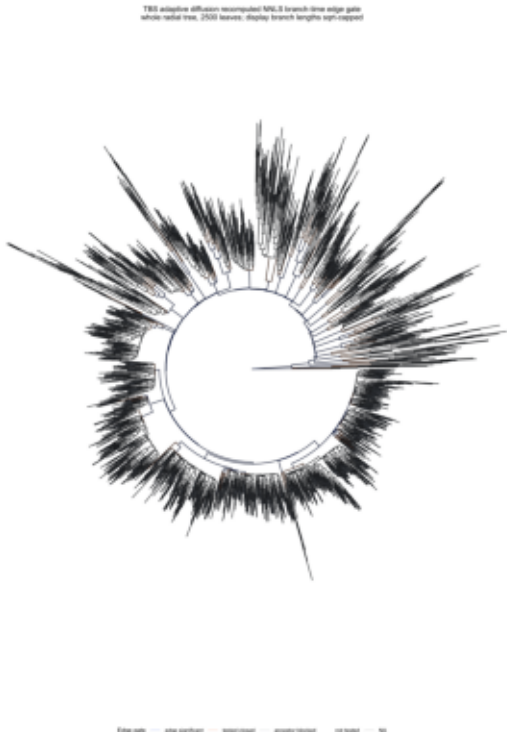
all 1,465 leaves; radius = cumulative NNLS branch time; colors match the labeled UMAP

Adult pancreas: selected NNLS-TBS tree diagnostics

Selected cluster radial tree

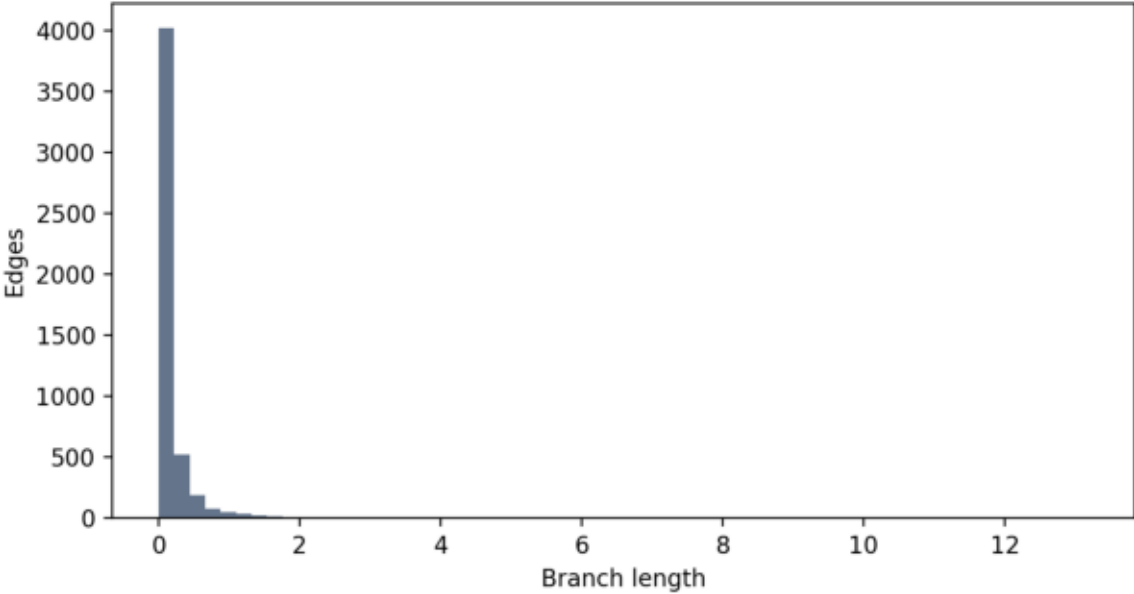


Selected full radial tree

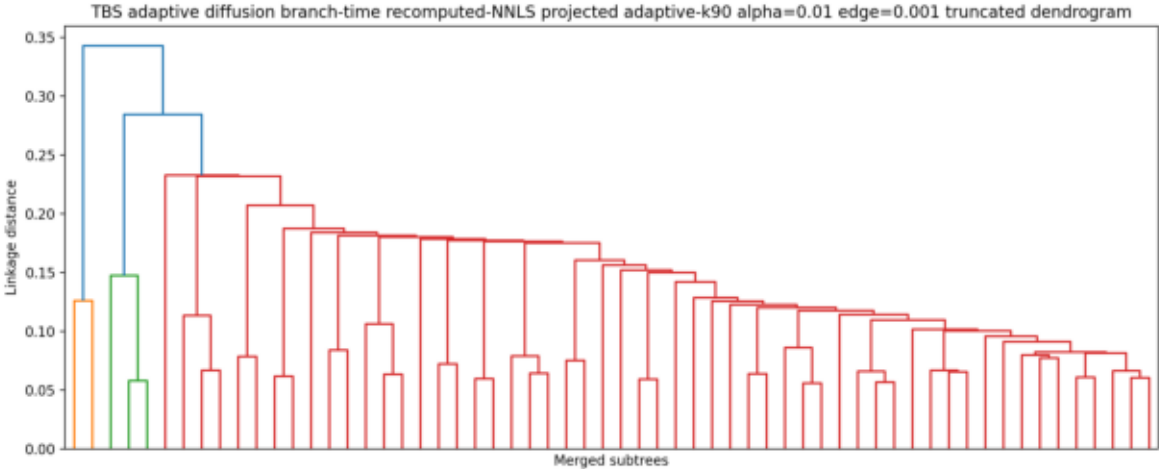


Selected NNLS branch lengths

TBS adaptive diffusion branch-time recomputed-NNLS projected adaptive-k90 alpha=0.01

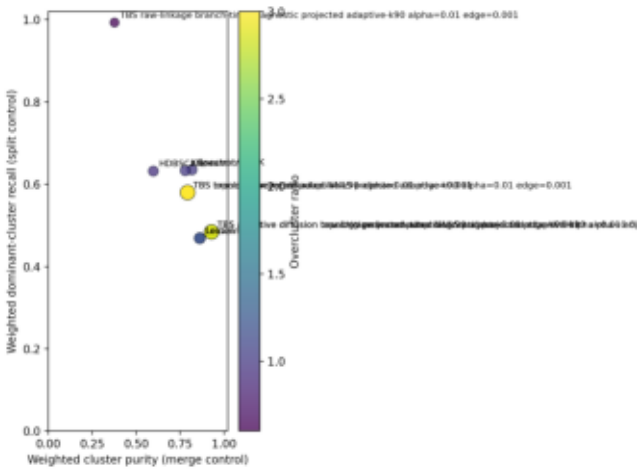


Selected dendrogram

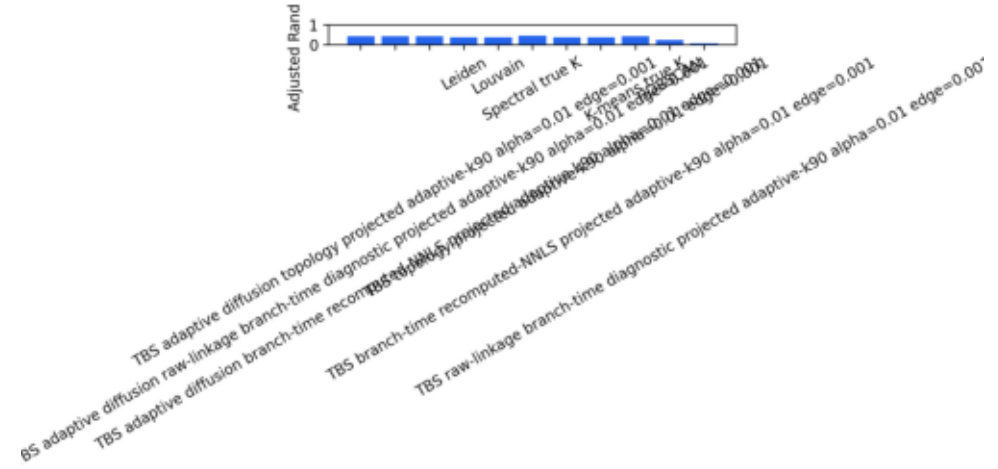


Adult pancreas: clustering and progenitor context

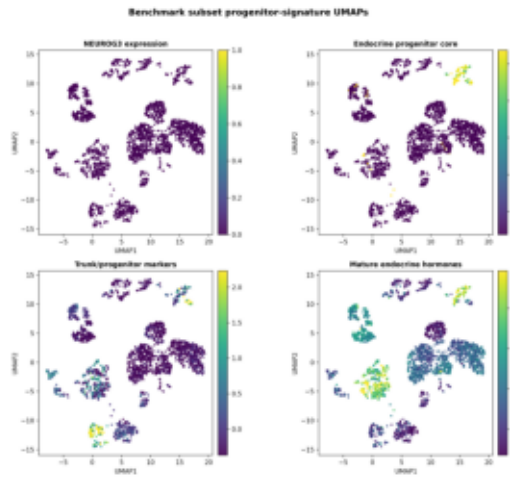
Method split/merge diagnostic



ARI benchmark summary

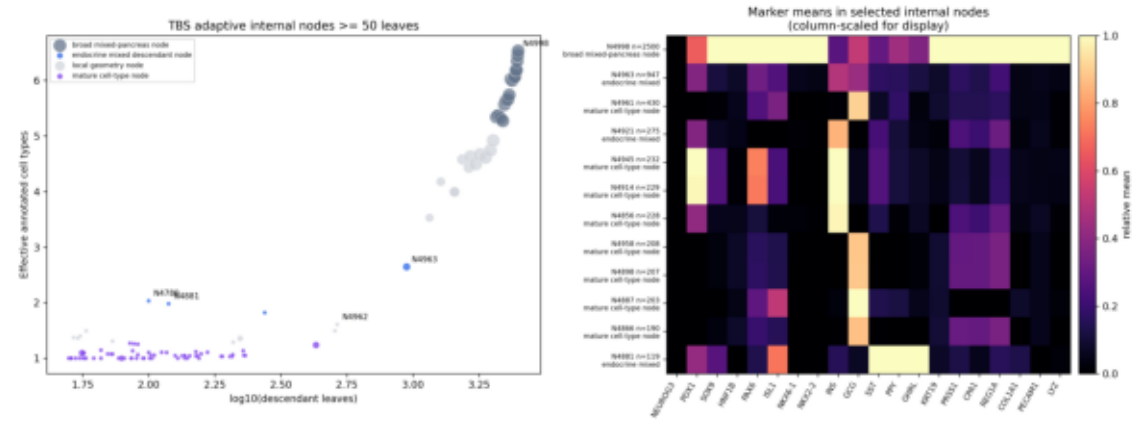


Progenitor signature UMAP

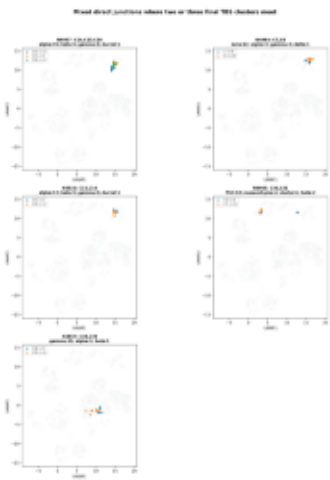


Internal-node progenitor comparison

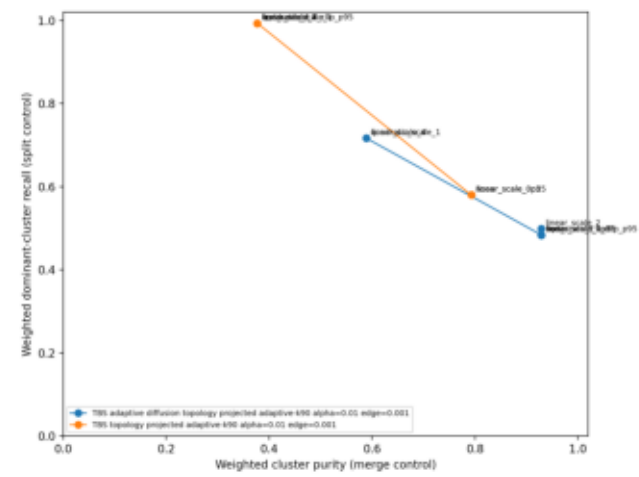
Inner nodes compared to progenitor-style lineage expectations



Mixed two/three cluster junction UMAP

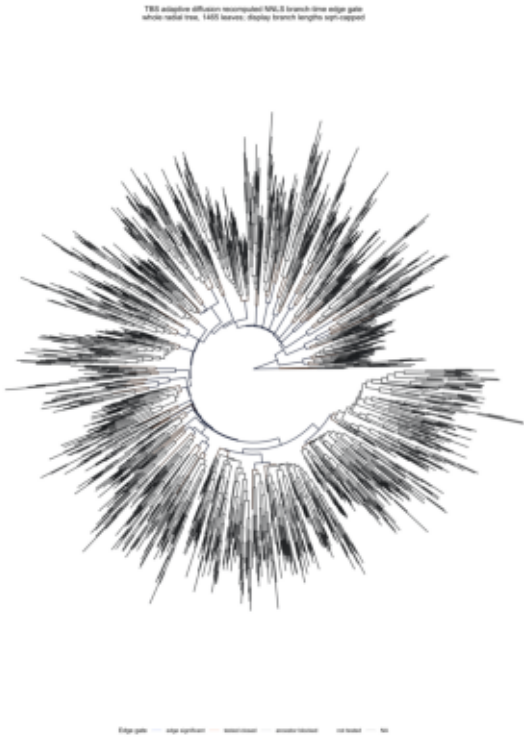


Branch-time sensitivity

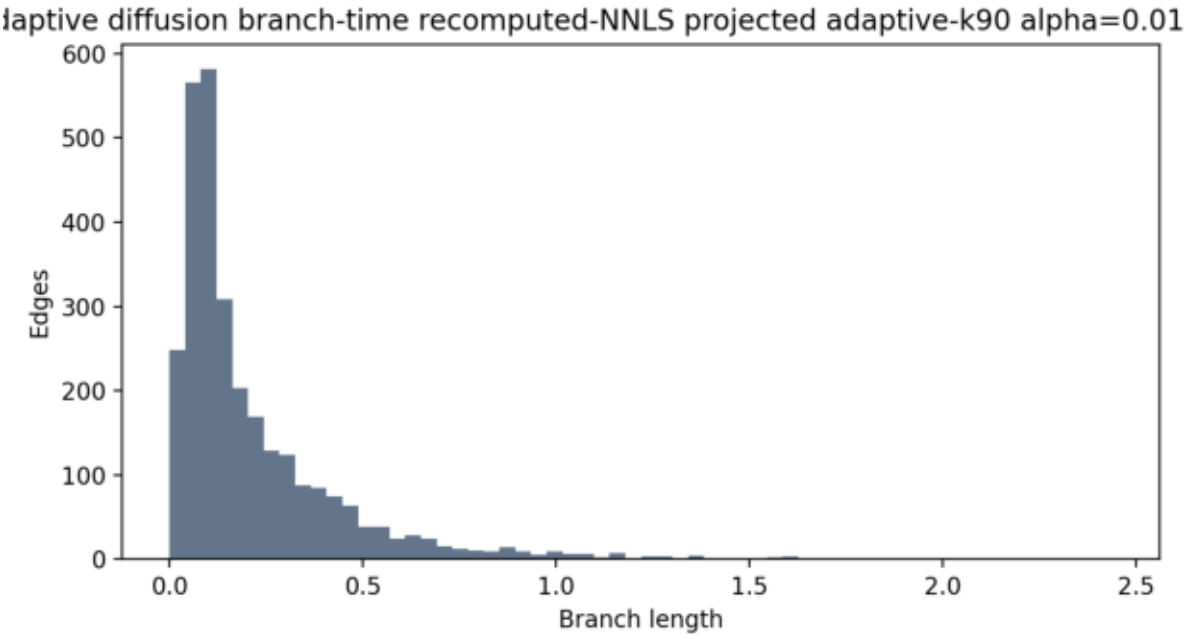


Goncalves: selected NNLS-TBS tree diagnostics

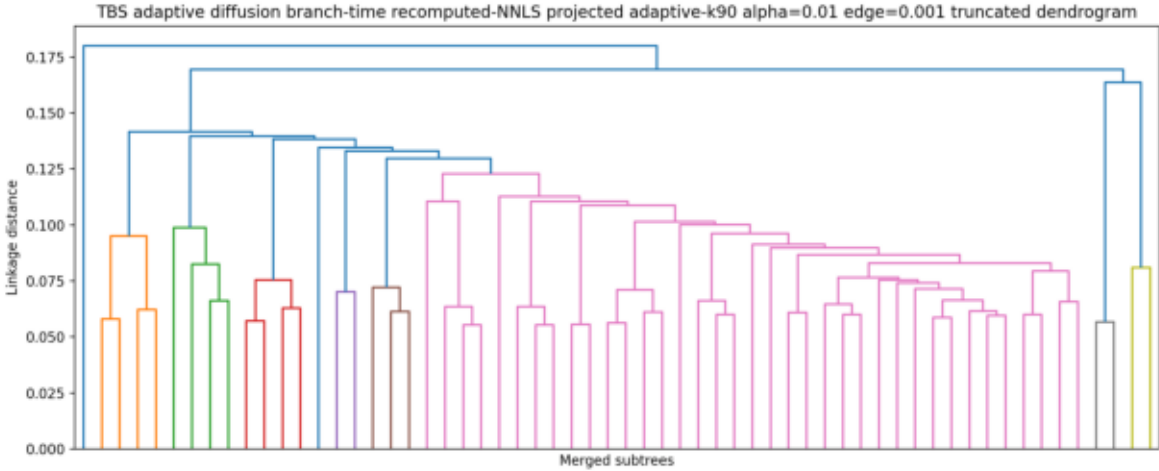
Selected radial tree



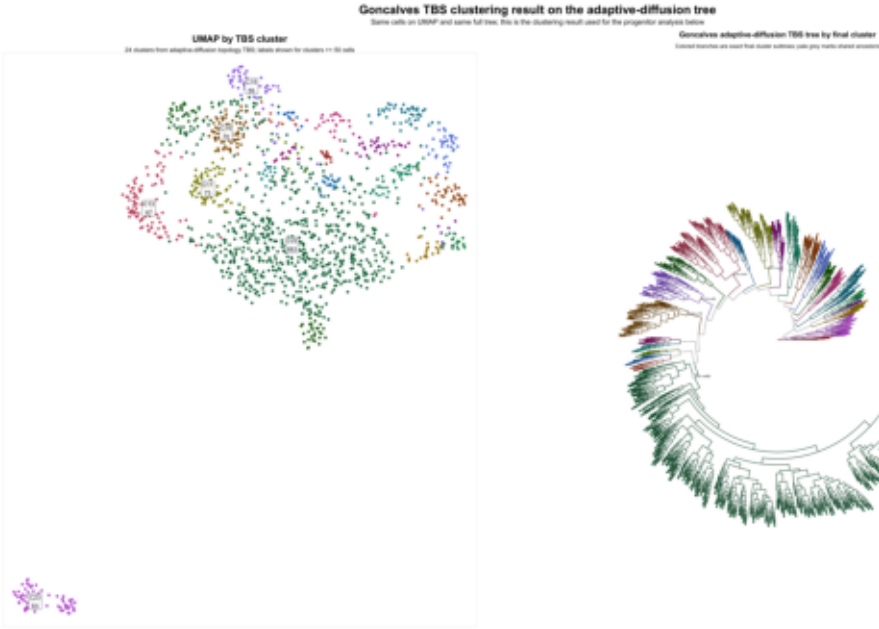
Selected NNLS branch lengths



Selected dendrogram



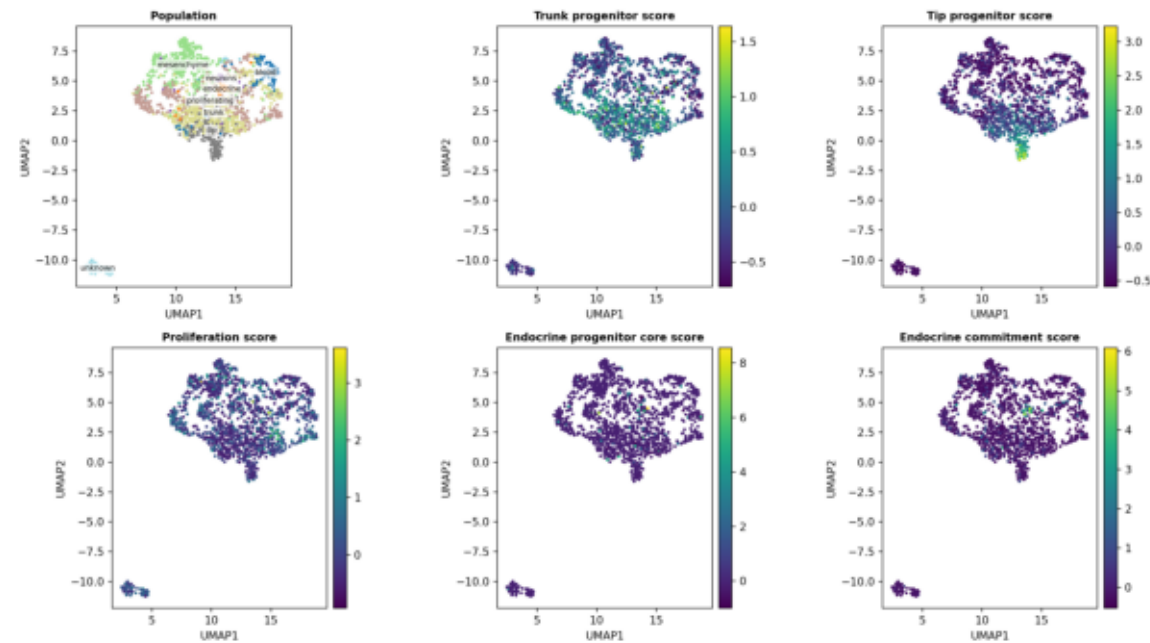
Cluster UMAP + tree page



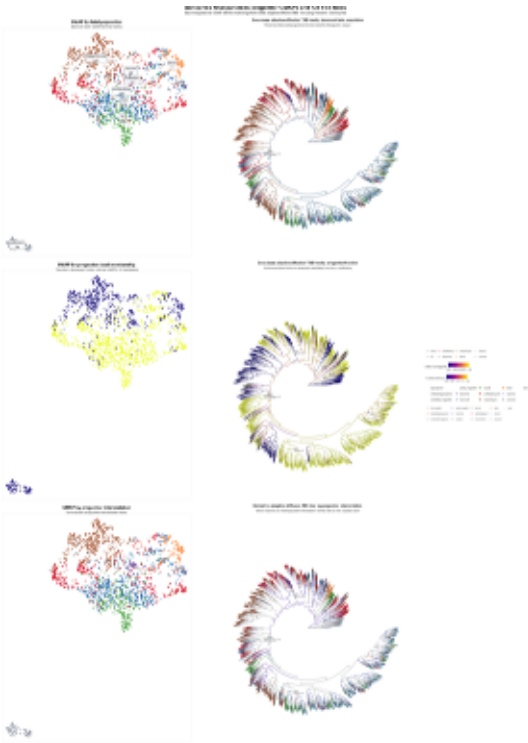
Goncalves progenitor UMAP and tree panels

Progenitor signature UMAP

Goncalves fetal pancreas progenitor signatures

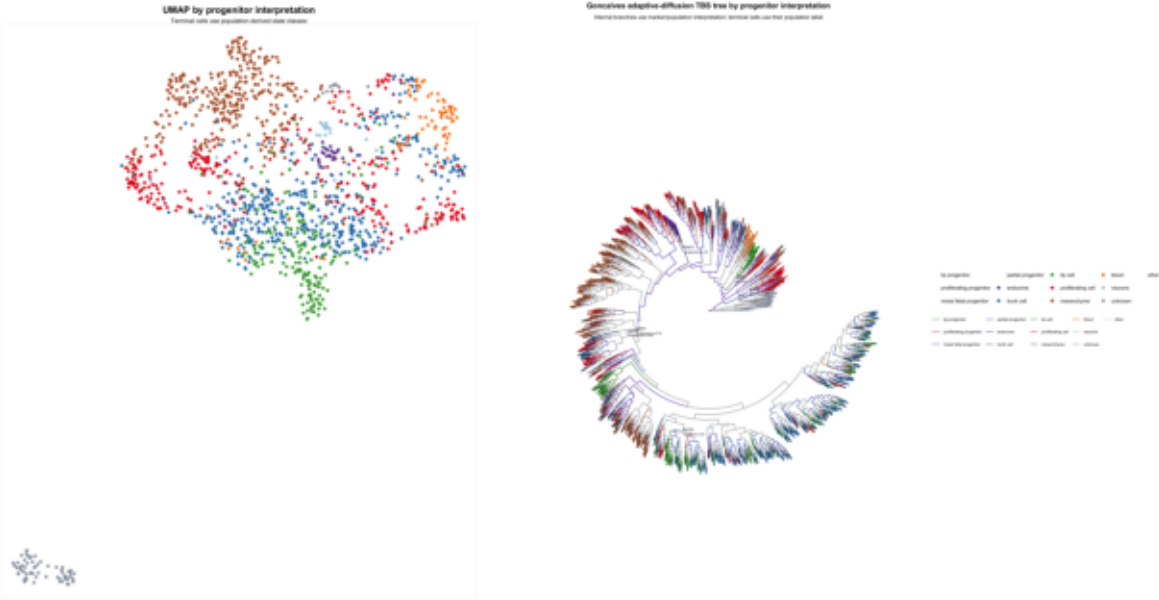


Progenitor UMAP + tree panel



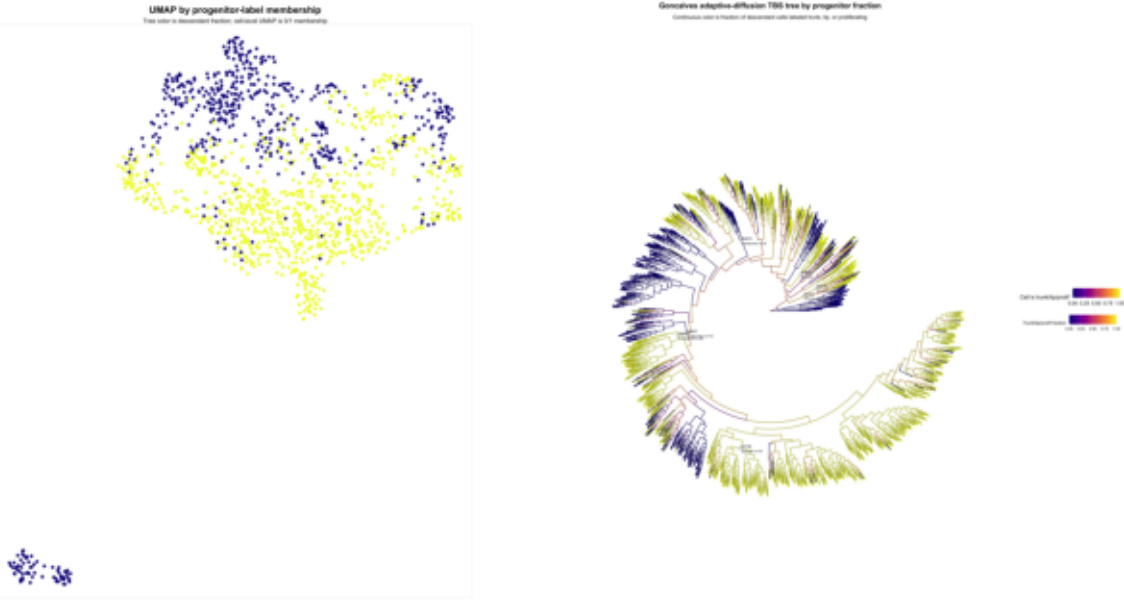
Progenitor state UMAP + tree

Goncalves progenitor-state UMAP and full TBS tree



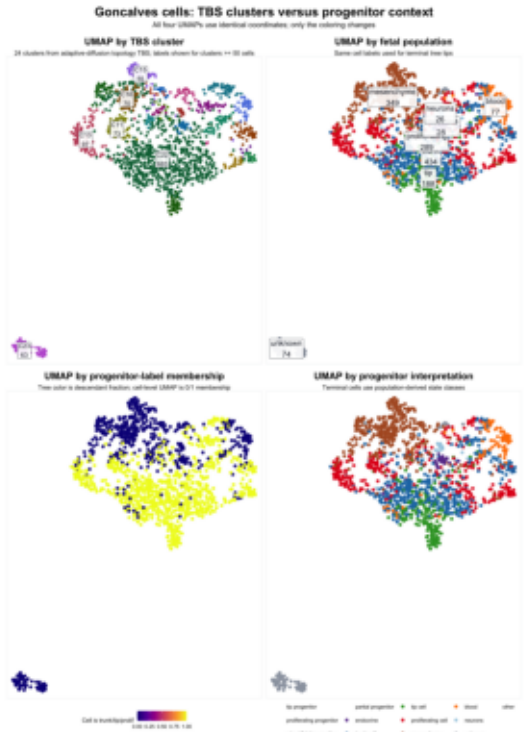
Progenitor fraction UMAP + tree

Goncalves progenitor-fraction UMAP and full TBS tree



Goncalves relationship panels

Relation UMAP grid

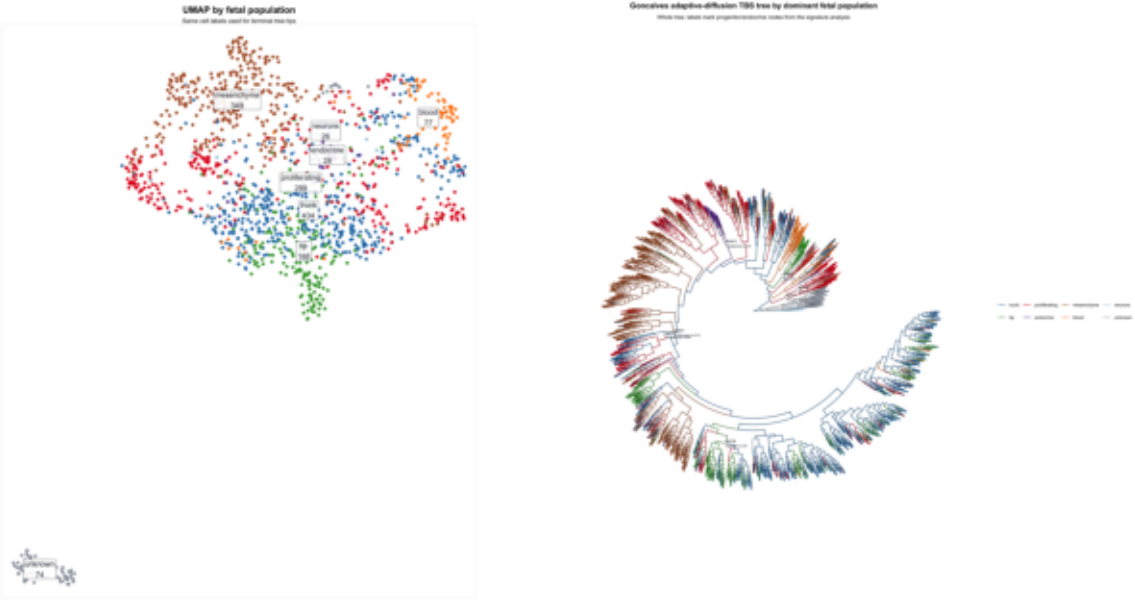


Relation tree grid

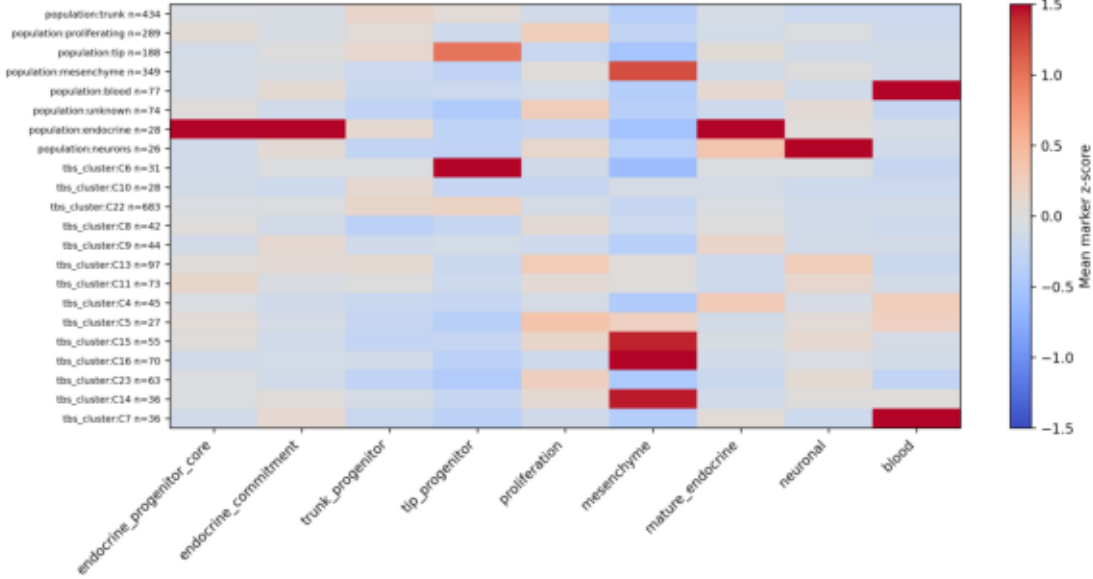


Population UMAP + tree

Goncalves fetal population UMAP and full TBS tree

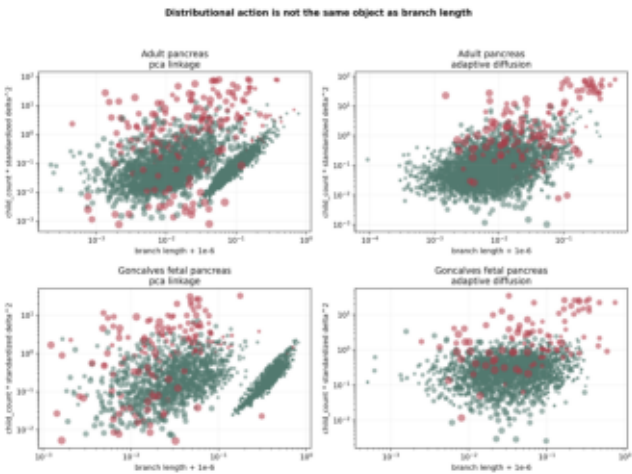


Population cluster signature heatmap

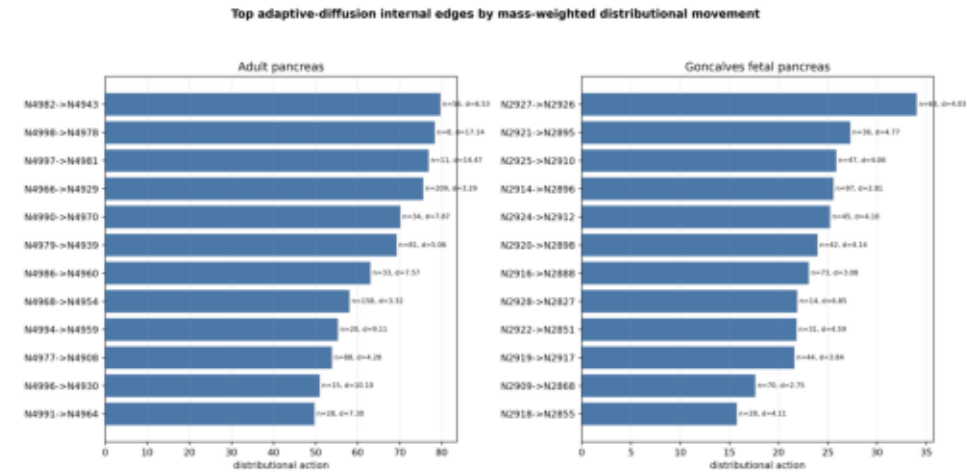


Branch-length and distributional-action audits

Distributional action vs branch length

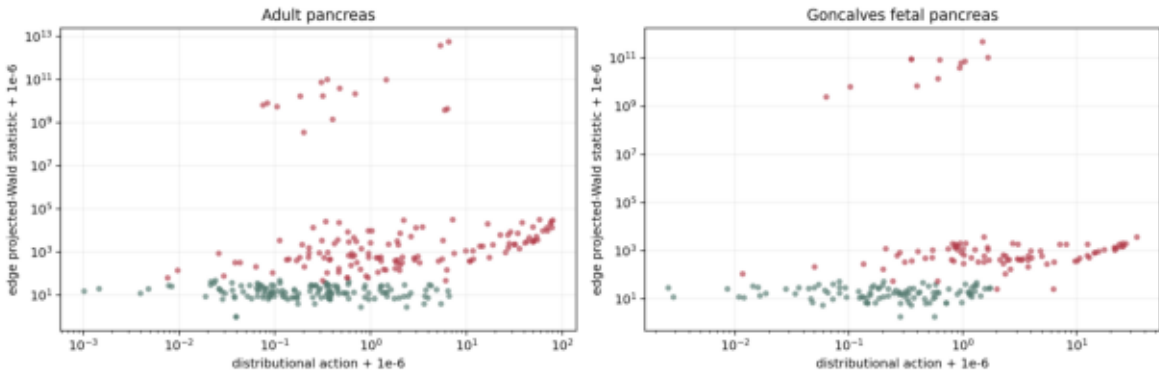


Top internal distributional-action edges

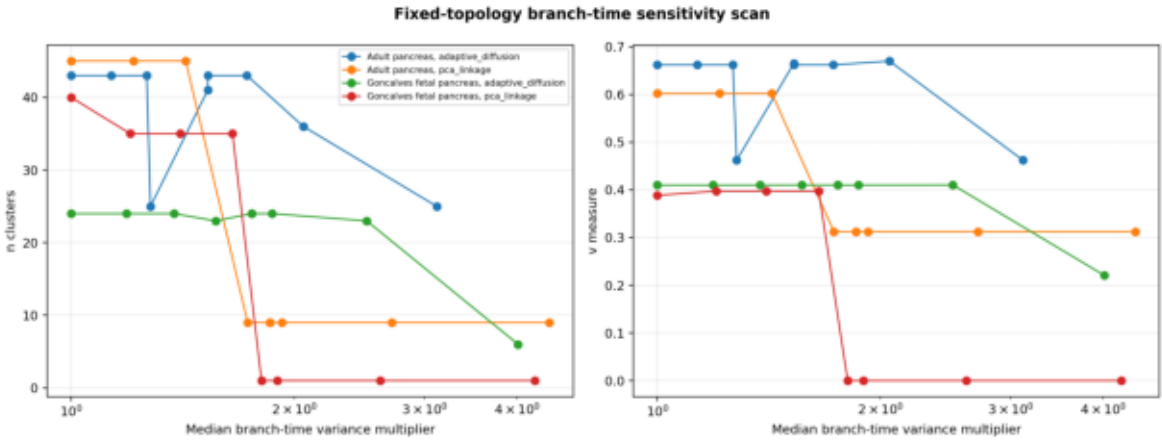


Distributional action vs edge statistic

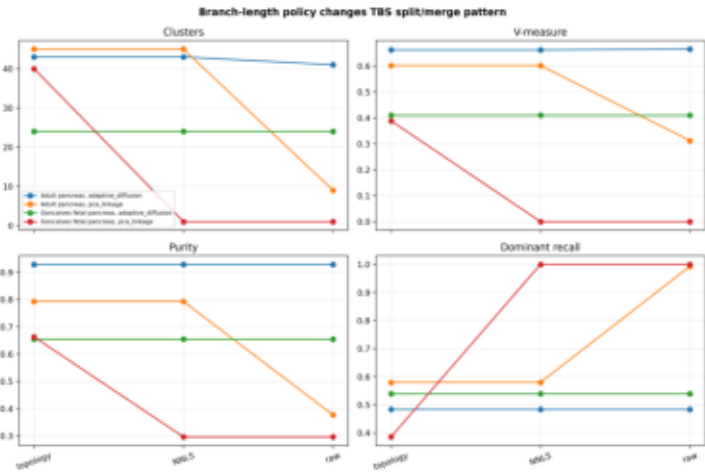
Mass-weighted movement tracks, but does not duplicate, edge-test evidence



Branch-time sensitivity effects



Branch-length cluster effects



Assignment similarity heatmap

